



AnimatLab: A 3D graphics environment for neuromechanical simulations

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ABSTRACT

The nervous systems of animals evolved to exert dynamic control of behavior in response to the needs of the animal and changing signals from the environment. To understand the mechanisms of dynamic control requires a means of predicting how individual neural and body elements will interact to produce the performance of the entire system. AnimatLab is a software tool that provides an approach to this problem through computer simulation. AnimatLab enables a computational model of an animal's body to be constructed from simple building blocks, situated in a virtual 3D world subject to the laws of physics, and controlled by the activity of a multicellular, multicompartiment neural circuit. Sensor receptors on the body surface and inside the body respond to external and internal signals and then excite central neurons, while motor neurons activate Hill muscle models that span the joints and generate movement. AnimatLab provides a common neuromechanical simulation environment in which to construct and test models of any skeletal animal, vertebrate or invertebrate. The use of AnimatLab is demonstrated in a neuromechanical simulation of human arm flexion and the myotactic and contact-withdrawal reflexes.

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1. Introduction

A major goal of neuroscience is to understand how the nervous system is organized to control behavior. The nervous system gathers sensory information about the body's relationship to the world, and then makes decisions and issues motor commands which change that relationship. The dynamics of the interaction among the central nervous system, the body, and the world are central to the functional control of behavior (Chiel and Beer, 1997). To govern behavior correctly, the nervous system must both predict and respond to the consequences of the animal's own movements and behavior, and do so on a millisecond to second time scale.

Despite many experimental successes in the analysis of the nervous system and behavior, the dynamic relationship between nervous function and the body is poorly understood. The kinematics and dynamics of many behaviors have been described, and the neural circuitry for some of these behaviors has been mapped in anesthetized or restrained animals, or in isolated tissue preparations. However, we lack a means of predicting how the function

and behavior of individual neural and body elements will affect the performance of the entire system and the behavior of the animal.

Neuromechanical simulation provides a new and promising approach to this problem. Computational models of the relevant neural circuits, body parts, and the physical world simulate the neural and biomechanical mechanisms of a behavior simultaneously in a physically accurate environment (Pearson et al., 2006). Here we introduce AnimatLab, a free, open-source, Windows®-based software tool written to provide a general simulator for neuromechanical processes of skeletal animals, both vertebrate and invertebrate. We first describe AnimatLab, its components and how they are used to construct models of the body and nervous system that interact to produce behavior in a virtual physical world. We then illustrate its use by presenting a model of spinal control of human arm flexion, the myotactic reflex, and the contact-avoidance reflex.

AnimatLab allows users to build neural circuit and biomechanical body models in a virtual physical environment, and then to record time-series of any variable(s) while viewing an interactive, 3D animation of the simulated behavior. Separate editors are used to build a model's body and nervous system, the objects and parameters that describe the surrounding physical world, and the patterns of physical and neural stimuli to be delivered to the model. Its interactive graphical user interface (GUI) permits users to view and

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interact with the model during the simulation, and to record and display data that describe its behavior. Both model construction and the simulations are carried out in an integrated environment. Construction of neural network models and 3D biomechanical body models use the point-and-click GUI in a way similar to professional CAD tools or 3D modeling tools like Maya or 3DS Max (www.usa.autodesk.com). Simulations and modules can be readily shared between investigators to allow others to examine and extend a simulation.

AnimatLab is available at www.AnimatLab.com with 45 video tutorials, and over 100 pages of help files that will guide new users in a click-by-click fashion through model construction and use of the program's different capabilities. AnimatLab requires no programming knowledge to use, but does assume a familiarity with neural and muscle physiology at the upper undergraduate/beginning graduate level. For those interested in adding new capabilities, AnimatLab's modular architecture makes it highly extensible via its pluggable, C++-based interface.

2. Methods

2.1. AnimatLab organization

AnimatLab has three interactive components: the GUI that enables model building and data graphing, coupled solvers for the neural circuit and biomechanics simulations, and an interactive 3D animation of the model's behavior in a virtual Newtonian world. In the model building portion, a "Project Workspace" (Fig. 1a) sets environmental parameters and provides access to editors for all the objects in the simulation: the nervous system and body of each animal, the ground and water surfaces, and all the fixed objects in the simulated environment. The "Body Plan Editor" (Fig. 1b) allows the user to assemble each model animal's body in a point-and-click, Lego®-like fashion from a variety of different part types. The "Behavior Editor" (Fig. 1c) allows users to construct the neurons and neural circuits that control the behavior of the model organism in a similar drag-and-drop fashion. Links are established between common elements (e.g., muscles, sensors) in the two editors, and the simulation is run by simultaneously operating and interacting solvers, the Vortex® simulator from CM Labs (<http://www.vxsim.com/>) for the biomechanics and a custom-made solver for neural interactions that uses an exponential Euler method (MacGregor, 1987) that is also used by GENESIS (Bower and Beerman, 2007). New neural solver plug-ins can be added by users, and each one can operate with a different integration time step from the others and from the physics engine. The model animal's movements in the virtual environment are under neural control as it responds to simulated physical and experimental stimuli. The autonomous behavior of the model is displayed graphically in a 3D animation (Fig. 1d) as the simulation runs, together with plots of the time-series responses of any designated set of neural or physical variables (e.g., membrane potentials, muscle forces and spatial displacements; Fig. 1e). Both the 3D animation and the data time-series can be recorded for off-line analysis, and the model parameter space can be explored by running multiple simulations with different parameter values simultaneously on different nodes of a grid computer.

2.2. Assembly of a body model

To build an animal model, users define rigid body parts in the Body Plan Editor (Fig. 1b), connect them with standard joints, enable them to move with actuators, and provide them with sensors to detect the environment. The connectivity of parts in the model is represented in a tree diagram (Fig. 1b, left). Several dif-

ferent types of body structures are currently available, including boxes, spheres, cones, cylinders, and polygon mesh models. A polygon mesh is a set of vertices and triangular faces that define the shape and volume for that part. All parts are assumed to have a uniform density, and the distribution of mass throughout the mesh volume determines the moment of inertia for that part. Meshes allow animal models to have more realistic structures, both visually and dynamically, by providing a more accurate representation of the body than is possible with simple geometric shapes. A wide array of mesh representations of animals' bodies can be purchased inexpensively from various graphics websites, including the human model presented (from www.exchange3d.com). Alternatively, a polygon mesh can also be obtained from a 3D laser scan of an animal's body using a computer-driven scanner (e.g., <https://www.nextengine.com>). Depending on the size of the animal and the scanner, the body can be scanned whole or as separate parts. For the body mesh to be used within AnimatLab, the body segments must be split into separate mesh files so that each can act as a single rigid body. For example, an arm would be split at the elbow to allow the upper and lower arms to be simulated with a joint between them. This reconstruction of the mesh file requires use of a 3D graphics program like Blender (www.blender.org). The separate body segment meshes are individually imported into AnimatLab and linked by the appropriate hinges to recreate the body with its moveable joints. Models of a human body, cat skeleton, and locust body are available at www.animatlab.com.

Rigid body parts have user-specified physical properties including dimensions, density, and drag; actuators include muscles, muscle spindles, motors, and springs; sensors include receptors for stretch, touch, odors, and tastes. Each body part is selected from a drop-down box and then placed, oriented, scaled, and shaped with a mouse in a 3D GUI. Joints are modeled as movement constraints that prevent any motion of the connected body parts that is not allowed by that joint type. The currently available joint types are planar hinge, ball and socket, one-dimensional sliding (prismatic), and fixed.

A fill-in "Properties" table (Fig. 1b, bottom left) can be used for exact placements and specifications of body parts and joints. A "description" cell in each properties table permits entry of text describing the object, or references to sources of the parameter values, or hyperlinks to those sources. Documentation of this sort is essential to distinguish between parameters and features based on experimental measurements from those that are made up to enable the model to work. This feature enables the model to be used as a knowledge base for the animal's neural circuitry and body structure.

Sensory receptors for touch, for stretch of a muscle spindle, and for chemosensory stimuli are implemented in AnimatLab (photoreceptors and auditory receptors are planned for a future version). Representations of each sensory receptor are created in both the body model and the neural circuit model to map the field of sensory receptors onto the population of sensory neurons. In the body model, single mechano- or chemoreceptors, or an array of such receptors, can be placed on the body surface where contact with an appropriate stimulus will activate it (Fig. 2a). Each mechanoreceptor has a 2D receptive field that describes its sensitivity to stimuli applied on the body surface in its vicinity. The distance from the contact to the center of the field is used to scale the force of the contact using a Gaussian function (Fig. 2a-2). In the neural circuit model, the corresponding representation of a body receptor is linked through a transduction adapter to a neuron compartment that represents the sensory neuron excited by the receptor. The scaled force produced at the receptor is transduced by a sensory adapter into a generator current (Fig. 2a-3) and passed into the sensory neuron (Fig. 2a-4). Receptors with overlapping receptive fields may detect the same physical contact and evoke spike responses in

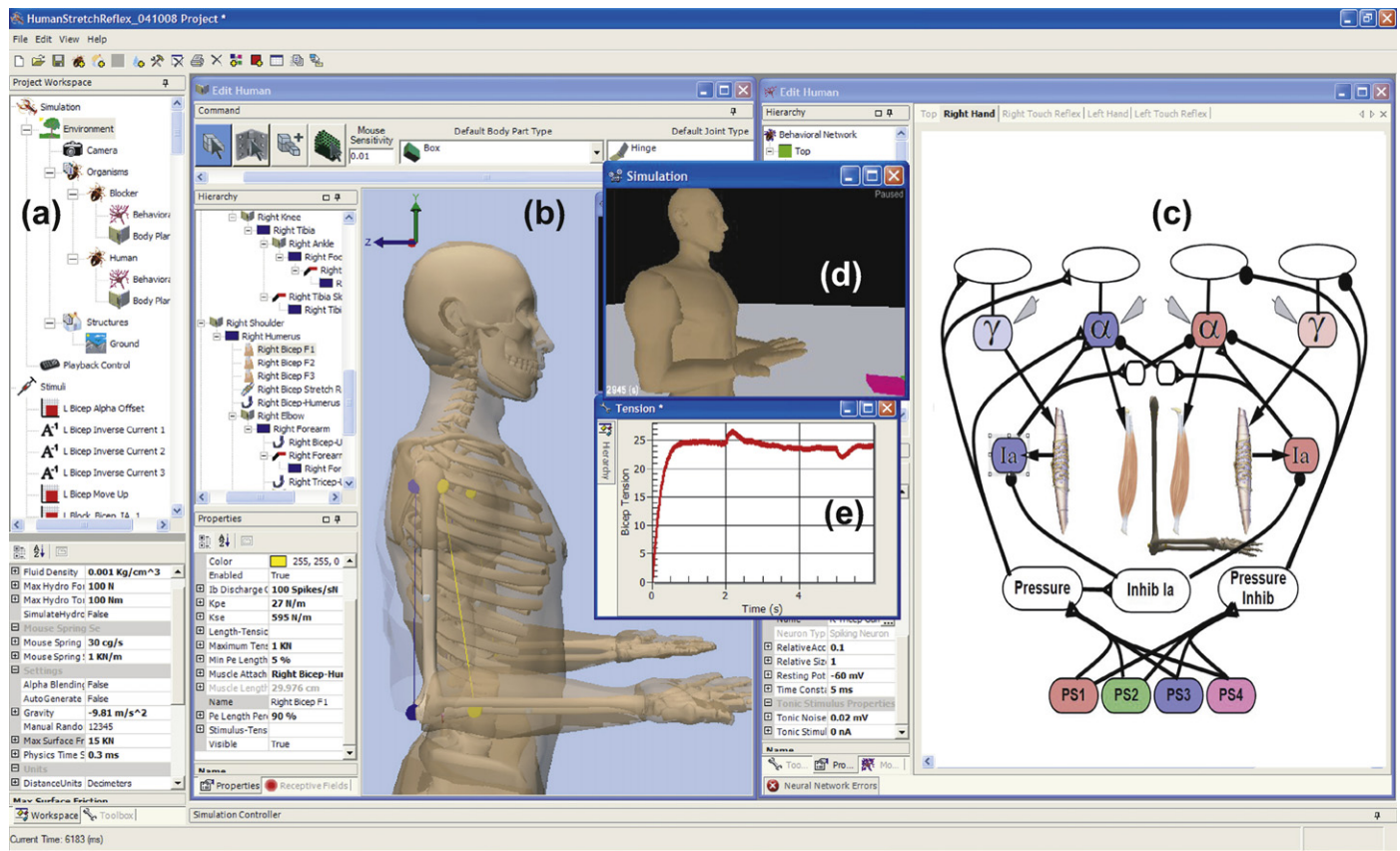


Fig. 1. Screenshot from AnimatLab showing the different editing and simulation components of the program. (a) The Project Workspace panel where users manage all simulation properties and components, as well as playback, stimuli, and data plots. (b) The Body Plan Editor that allows users to build organisms in a point-and-click manner. The top bar of this window lets users switch between viewing rigid bodies, joints and receptive fields, and it allows them to select the default body and joint types that are used when new parts are added. The panel in the top-left of this window displays the hierarchical connectivity of the body parts, and the property table below that lists all properties for an individual part. (c) The Behavior Editor allows users to drag-and-drop neurons, muscles, and other electrically excitable parts to create neural circuits, and then draw synaptic connections between these elements. Tabs at the top of the diagram list all the pages for the entire network, and the property panel on the bottom-left lists properties of the selected neuron. (d) The simulation window shows the 3D graphical display of the simulated movement. Users can use the mouse to alter the 3D view and manipulate objects during the simulation. (e) Data charts allow the user to plot a time-series of any set of parameters in the simulation.

their sensory neurons that reflect the different transduction current amplitudes. All physical processes, including joint rotation and extension of a stretch receptor, use this adapter transduction mechanism to produce a generator current that can be applied to a sensory neuron.

The biomechanics of muscles and muscle spindles (or their invertebrate analogs) are represented by Hill-based models (Shadmehr and Wise, 2005) that consist of a serial spring in series with the parallel combination of a parallel spring, a dashpot, and a force actuator (Fig. 1b, yellow biceps and blue triceps; Fig. 2b) (for interpretation of the references to color in this sentence, the reader is referred to the web version of the article). Users attach a muscle to the body by adding attachment points to body segments on either side of a joint, and then stringing the muscle between these points. Multiple attachment points can be used to allow the muscle to span multiple joints, or to control the direction of the applied force. Muscle model properties are determined by the resting muscle length, the spring and dashpot constants, the stimulus–tension curve, and the length–tension curve. Neural control of muscle tension is mediated by a motoneuron (MN) (Fig. 2b-1) that synapses onto a compartment that represents the common electrical properties of the muscle fibers that compose the muscle (Fig. 2b-2). A sigmoidal stimulus–tension curve relates the force level of the actuator to the muscle membrane potential (Fig. 2b-3). The actuator force is further scaled by the length–tension curve, an inverse parabola that determines the percentage of actuator force that is applied at a given muscle length

(Fig. 2b-4). The actuator force is then applied to the muscle to cause contractions (Fig. 2b-5).

A muscle spindle responds to both imposed and excited stretch through an adapter that translates muscle responses into a generator current that is passed into a sensory neuron. Multiple adapters can be used to allow sensory responses to be as complex as required. For example, a 1a afferent may respond to current produced by one adapter sensitive to the serial-elastic (SE) length of a muscle and to current produced by a second adapter sensitive to the rate of change of SE length.

Motors are another way that movements can be generated by models within AnimatLab. The hinge and prismatic joints can be motorized to produce controlled rotational and translational motion. This can be useful in simulating passive movements imposed on a limb during an experiment, or in substituting for patterns of muscular force generation around a joint when only joint dynamics are known.

2.3. Neuron and network models

In the Behavior Editor (Fig. 1c), single- or multicompartment neurons and neural circuits are constructed by dragging elements from a toolbox onto a circuit editing page, and linking them by electrical or chemical synapses. Available elements include spiking integrate-and-fire (I&F) compartments (Destexhe, 1997; Hines and Carnevale, 1997; MacGregor and Lewis, 1977), Hodgkin–Huxley (HH) compartments (called ‘Non-Spiking’ in the

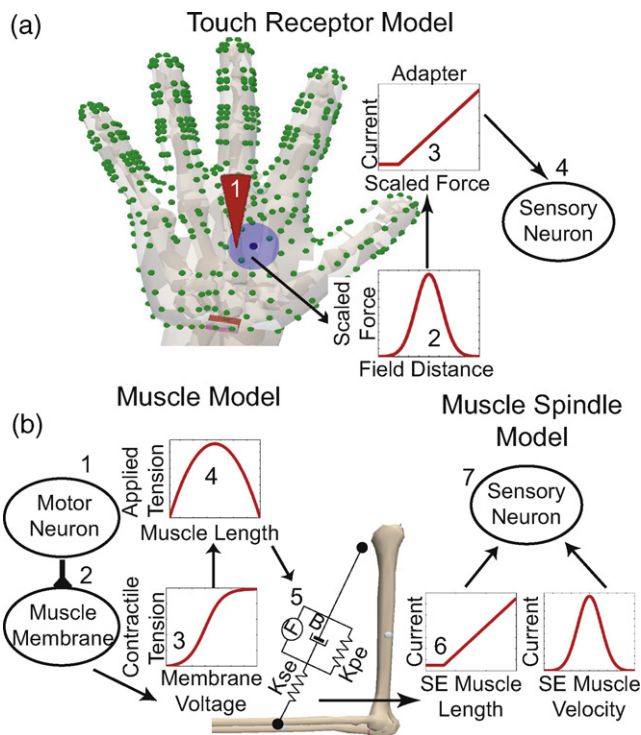


Fig. 2. Modeling of sensory and motor systems in AnimatLab. (a) Mechanosensory receptive fields are located over the surface of the hand. The center of each field is shown as a green dot. When an object contacts the hand (1), the field distance, or distance from the contact point to the center of the field, is calculated and used to scale the force of the contact (2). Scaled force is used by an adapter (3) to calculate a generator current that is applied to a sensory neuron (4). (b) Tension in a muscle or spindle is controlled by firing of one or more motor neurons (1), which depolarize the muscle membrane (2). The muscle membrane voltage is transduced into a contractile tension (3), which is proportionally scaled based on the current muscle length (4), and then applied at the force generator (F) in the Hill muscle model (5) to produce joint torques and movements. Stretch receptors use an identical model, but muscle properties like length and velocity are transduced into currents (6) that are applied to a sensory neuron (7).

program to distinguish them from I&F compartments) that include voltage- and time-dependent ionic conductances, and firing rate (FR) neurons (Beer, 1990; Vogels et al., 2005). Neurons can be modeled as single compartments or as multiple compartments linked by ohmic coupling conductances. All parameters of each compartment are displayed in a “Property Table” where they can be changed (Fig. 1c, bottom left). Both conductance-based and current-injection synapses are available and can be added to the network by connecting neuron elements with cursor-drawn lines. Users can define new instances of these synapse types as the need arises. More detailed descriptions of the neural models used by AnimatLab are available in the online documentation at www.animatlab.com/help.

The neural circuit can be enlarged by adding new circuit editing pages from the neural editor. Any individual compartment on one page can be represented on another page by an ‘Off-Page Connector’. The network can also be divided into hierarchical neural subsystems. This simplifies the organization and maintenance of the entire network and allows it to be split out into functional subsystems. For example, it is possible to create a sub-network that controls movement of a leg, and then copy, paste, and modify it for each of the remaining legs.

The different neuron compartment and synapse types make it possible to incorporate different levels of abstraction in the same model according to the question being addressed. For example, some neurons may be modeled as complex multicompartments with both HH and passive compartments, while other

parts of the simulated nervous system are modeled with single-compartment I&F neurons or FR models that each reflect the average firing rate responses of many similar neurons to a common input. Use of different levels of abstraction allows models to capture details of the living system that are needed to examine an hypothesis while allowing supportive aspects to be modeled more abstractly. This arrangement economizes on model complexity and on computation time needed for simulations.

2.4. Construction of the human arm and reflexes

Several examples of AnimatLab’s function are given at www.AnimatLab.com, including a hexapod walker, a grasshopper jumping, and a crayfish escape response. Here we describe a model of human arm flexion and the myotactic and avoidance reflexes that we created to demonstrate AnimatLab’s capabilities. The model is not a serious proposal for how these command responses and reflexes might work, but it is sufficiently detailed to show how a set of hypotheses about dynamic control can be examined and tested in a more complete model. The human model described below is available at www.AnimatLab.com/HumanMyotacticReflex.

A model of the human body was developed from a 3D polygon mesh that was purchased online (from www.exchange3d.com) and then broken into individual bones and body segments using the graphics program Blender (www.blender.org; Fig. 1b). Each bone and body segment was re-scaled to the appropriate dimensions to match published values (Clauser et al., 1969). All parts are assumed to have a uniform density, and the distribution of mass throughout the mesh volume determined the moment of inertia for that part. The model consisted of separate bone and non-skeletal elements, in which the bones were inside the non-skeletal elements. The density of all bones was set to 1.9 g/cm³ (Cameron et al., 1999), and the densities of the overlying non-skeletal segments (representing the skin, muscle mass, connective tissue and blood) were calculated by subtracting the mass and volume of the bone from the mass and volume of the overlapping body segment (Clauser et al., 1969), and calculating the resulting mass/volume ratio. The bone and non-skeletal elements of a limb segment (e.g., the forearm) were bound together in the model by a fixed joint.

Here the focus was on arm flexion at the elbow, so the upper arm and body were fixed in space, the forearm and hand rotated around a hinge joint at the elbow, and the hand was connected to the forearm by a fixed joint (Fig. 1a). The position, orientation, and angular limit of the elbow hinge joint was set so that only the normal range of arm movements could occur.

Forearm movement in the model is produced by contractions of a biceps flexor muscle and a triceps extensor muscle; reflex responses are mediated by a pair of corresponding muscle spindles. Values for the spring and dashpot coefficients of the biceps and triceps muscle and muscle spindle models were obtained from Massone and Myers (Massone and Myers, 1996; Myers and Massone, 1997). The measured maximum tension for these muscles (Myers and Massone, 1997) set the upper limits for the stimulus – tension curves, and the length – tension curve was assumed to follow measurements obtained for similar muscle types (McMahon, 1984).

2.5. Reflex circuitry

The model myotactic stretch reflex circuit (Fig. 3a) (Pearson and Gordon, 2000; Windhorst, 2007) enables the arm to resist perturbations away from its equilibrium position. The circuit contains descending tonic command neurons (FR models) that drive the alpha and gamma MNs (I&F models); they, in turn, excite the working and spindle muscles, respectively. Spindle tension generates depolarizing membrane current in a 1a sensory neuron (I&F) that

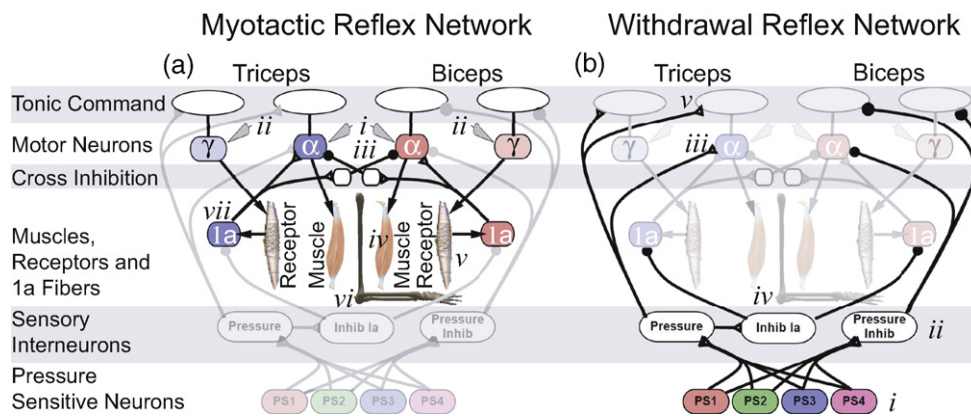


Fig. 3. Model neural circuits controlling movements of the arm. (a) Tonic commands to alpha and gamma motor neurons (MNs) set the tonic MN activity level to obtain the initial arm equilibrium position. The myotactic reflex network mediates responses to deviations from the planned movement of the arm. To initiate arm flexion, depolarizing and hyperpolarizing currents (i) drive flexor and extensor alpha MNs (iii), respectively, to cause biceps contraction and triceps relaxation (iv), while gamma MNs are stimulated (ii) to maintain tension in the muscle spindles (v). Tensions in muscles produce joint torque and movement of the arm (vi). Responses of the 1a sensory neurons (vii) signal an error in the trajectory of a movement. (b) The withdrawal reflex mediates responses to noxious contacts. The reflex is activated by strong contacts on the palm of the hand that excite pressure-sensitive neurons PS1–4 (i). The sensory neurons excite local interneurons and inhibit the myotactic reflex (ii) in a fast spinal feedback loop (iii) to move the hand away from the contacting stimulus (iv), while a slower central feedback loop changes the descending tonic commands (v) to produce a new limb posture.

may fire to provide afferent feedback to the alpha MNs and (I&F) inhibitory interneurons. The inhibitory neurons inhibit the antagonist alpha MNs.

The model withdrawal reflex (Fig. 3b) (Pearson and Gordon, 2000) extends the arm rapidly away from a strong, noxious contact on the upturned palm of the hand and causes the limb to adopt a new posture away from the stimulus. The stimulus falls within the receptive fields of four model mechanosensory neurons (Figs. 2a and 3b: PS1–4) that each respond in proportion to the amount of force they experience. These sensory neurons excite a central interneuron (Fig. 3b, “Pressure”) that excites the extensor MNs and inhibitory interneurons of both the flexor MNs and the stretch reflex for both muscles. This promotes extension of the arm and inhibits the stretch reflex. This is the spinal feedback loop. In the longer central feedback loop, the interneurons ascend to the tonic command neurons to alter their firing rate.

3. Results

A voluntary arm flexion was simulated by command current activation of the MNs. At the outset, the tonic activity of the alpha MNs caused the flexor (bicep) and extensor (tricep) muscles to maintain constant tensions that kept the forearm at a 90° angle relative to the upper arm (Fig. 4a–1). A step excitatory command current was applied at 1.0 s to the biceps alpha MNs to simulate an arm flexion motor command (Fig. 4b). The brief rise in flexor tension caused the arm to flex rapidly at first, and then more slowly to adopt a new, flexed, equilibrium position. When the command current was turned off, the equilibrium position reverted to the 90° angle and the arm returned to that position. To produce a faster movement, a multiphase pattern of current stimuli evoked a corresponding pattern of flexor and extensor MN activity that started the arm to flex quickly, then slowed the flexion, and finally maintained the arm at the new flexed equilibrium position (Windhorst, 2007) (Fig. 4c). This pattern of stimulation produced a smooth movement of the arm to the target position in about half a second. The relative amplitude and timing of each phase of the motor command proved to be critical for a smooth, rapid movement; deviation of more than a few percent in the amplitude or timing of any phase of the pattern produced over- or undershoot of the flexion or re-extension movement (not shown).

Command currents were also passed to the flexor and extensor gamma MNs to keep spindle tension constant during the movement. The biceps shortened during arm flexion and the triceps

were stretched, which reduced the tension in the flexor spindle and increased it in the extensor spindle. One hypothesis of 1a reafference is that efferent signals to the spindles maintain the sensitivity of the receptors at a base level during planned movements. This keeps the 1a feedback constant unless an external perturbation or unexpected load does not permit the spindle to contract as planned, so that spindle tension rises unexpectedly. Deviations from the base tension levels are interpreted as errors from the planned movement and evoke changes in 1a afferent activity that excite resistance reflexes to restore the spindle tension (Shadmehr and Wise, 2005). In this simulation, increased excitation of the flexor gamma MN during arm flexion helped maintain the flexor spindle tension, while decreased excitation of the extensor gamma MN maintained the extensor spindle tension (Fig. 4b and c). When the arm was moved without gamma co-activation, the changes in spindle tension excited the myotactic stretch reflex, which attempted to return the arm to its original position. This is illustrated in Fig. 4d, where a three-phase command current produced arm flexion, but the gamma co-activation was disabled so that gamma MNs received only tonic excitation. The tension in both the biceps and triceps spindles varied widely and oppositely from the desired constant levels, which led to excitation of the flexor 1a afferents during flexion, and the extensor 1a afferents during re-extension. These 1a responses excited MNs to oppose the movements, which became jerky and failed to reach the flexion target.

The myotactic reflex was activated when the hand encountered a small, heavy block in its path as the arm flexed in response to the tri-phasic command (Fig. 5). The block was on a hinge joint that allowed it to move out of the way if the arm produced sufficient force. The arm flexed until it hit the block, which momentarily stopped the movement. The resulting deviation from the planned motion produced a rise in the biceps spindle tension when the interrupted flexion prevented the spindle from shortening in response to its gamma MN command (Fig. 5b, Receptor Tension). The increased flexor spindle tension excited the 1a afferent, which added to the flexor alpha MN excitation. The extra flexor force was sufficient to push the arm past the block and overshoot the goal position. Additional reflex activity allowed the arm to settle to its target position.

The contact-withdrawal reflex was activated when the hand encountered a sharp spike placed in the path of the hand’s movement during arm flexion (Fig. 6a). Four sensory receptors with overlapping pressure-sensitive receptive fields in the palm of the hand (Fig. 6b) responded to the contact by producing generator

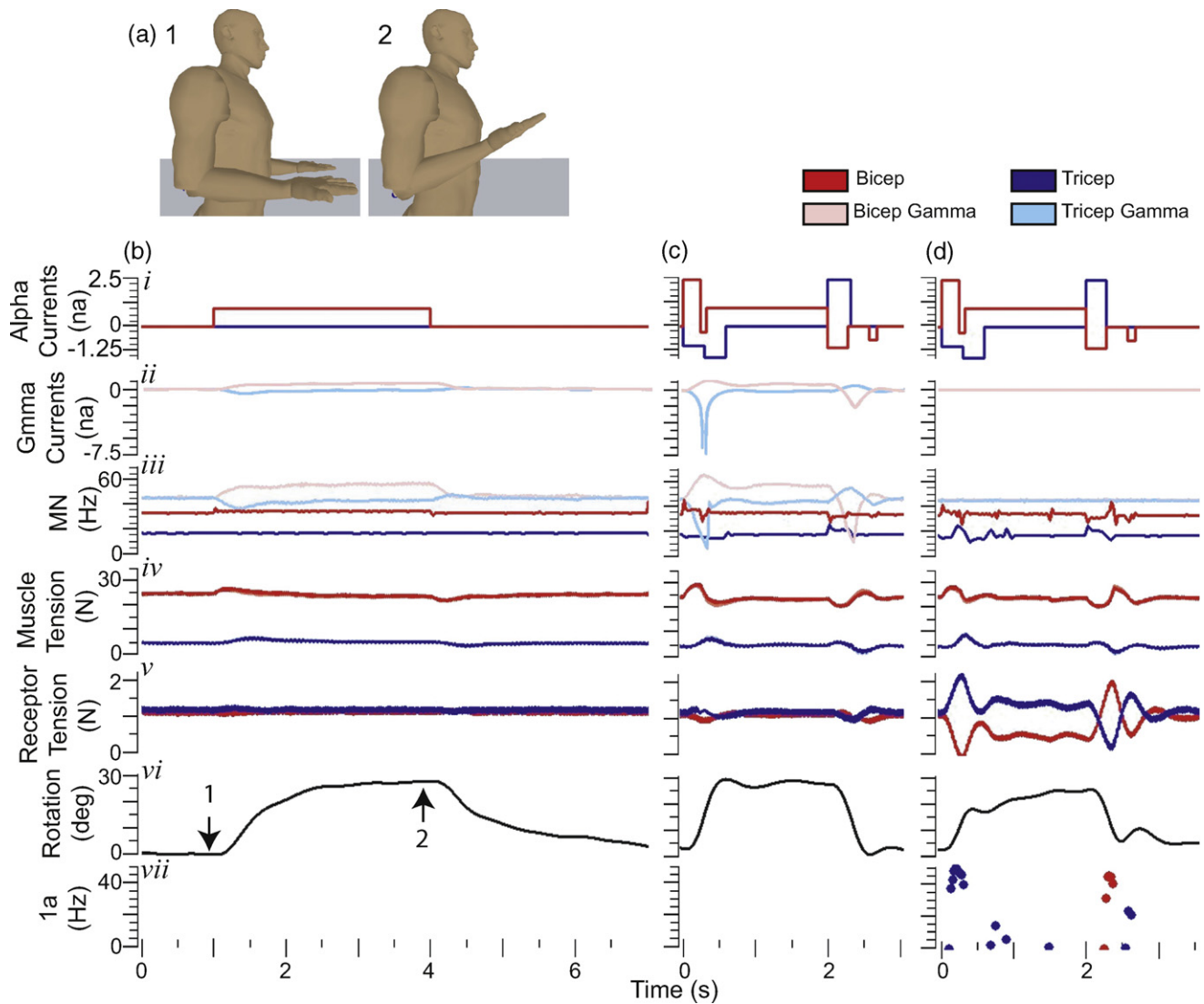


Fig. 4. Simulation of voluntary arm flexion. (a) Screenshots of the animated arm movements from the beginning position (1) to the final position (2). (b–d) Plots of model variables; the italicized Roman numeral of each plot corresponds to a numbered item in the model diagram of Fig. 3a. (i) The current stimuli applied to biceps and triceps alpha MNs. (ii) The current stimuli applied to the gamma MNs. (iii) The MN firing frequencies. (iv) Tension of the biceps and triceps muscles. (v) Tension of the muscle spindles (Receptors). (vi) Elbow rotation. (vii) 1a frequency response. (b) Responses to a current step applied only to the biceps alpha MN and to depolarizing and hyperpolarizing currents applied to the gamma MNs of both muscles. Numbered arrows refer to the animations in (a). (c) Responses to a multiphasic command current pattern applied to the alpha and gamma MNs that produced a much faster movement by accelerating the arm quickly, braking its motion, and then maintaining the new position. (d) Responses to the same current pattern as in (c), but with gamma co-activation disabled.

currents in the corresponding sensory neurons (Fig. 6c). The circular receptive fields have bell-shaped spatial-sensitivity functions, so that the current produced in each cell depended on the magnitude of the force and the distance between the point of force application and the field center. In the example presented, pressure sensor 2 (Fig. 6b, PS2) produced the largest current, while the sensory receptors on either side produced less current; the firing frequencies of their neurons were proportional to these currents. The sensory neurons excited a set of central interneurons (Fig. 3b) that triggered the withdrawal response. Of necessity, the withdrawal response had two phases, an immediate 'spinal' phase, and a longer-latency, persistent phase. The sensory interneurons excited arm extensor MNs and inhibited arm flexors to reverse the direction of the hand movement away from the spike (Fig. 6a and c). By itself, this spinal reflex was insufficient, as the arm would try to return to its new, flexed equilibrium position as soon as contact with the spike had ceased. To prevent this, the sensory interneurons also acted on the higher tonic command neurons (Fig. 3b) to change the position of the equilibrium point away from the spike (Fig. 6a and c). A video

of all of the movements that have been described is available in supplemental material section as video 1.

4. Discussion

Neuromechanical simulations have been constructed to study the behavior of humans (Cheng et al., 2000; Delp and Loan, 2000; Taga, 1995a,b) and other animals, including insects (Cruse et al., 1998; Ekeberg et al., 2004; Spragna et al., 2007), lamprey (Ekeberg and Grillner, 1999), cat (Ekeberg and Pearson, 2005; Lockhart and Ting, 2007), and leech (Skierczynski et al., 1996). Some of these models are more biologically realistic (Ekeberg et al., 2004; Ekeberg and Grillner, 1999; Ekeberg and Pearson, 2005; Skierczynski et al., 1996; Taga, 1995b), while other models are more abstract (Cruse et al., 1995, 1998; Reichler and Delcomyn, 2000). These studies have shown that the simulation of the interacting neuronal and biomechanical components of the virtual animal can lead to valuable insights into the mechanisms that govern behavior, including

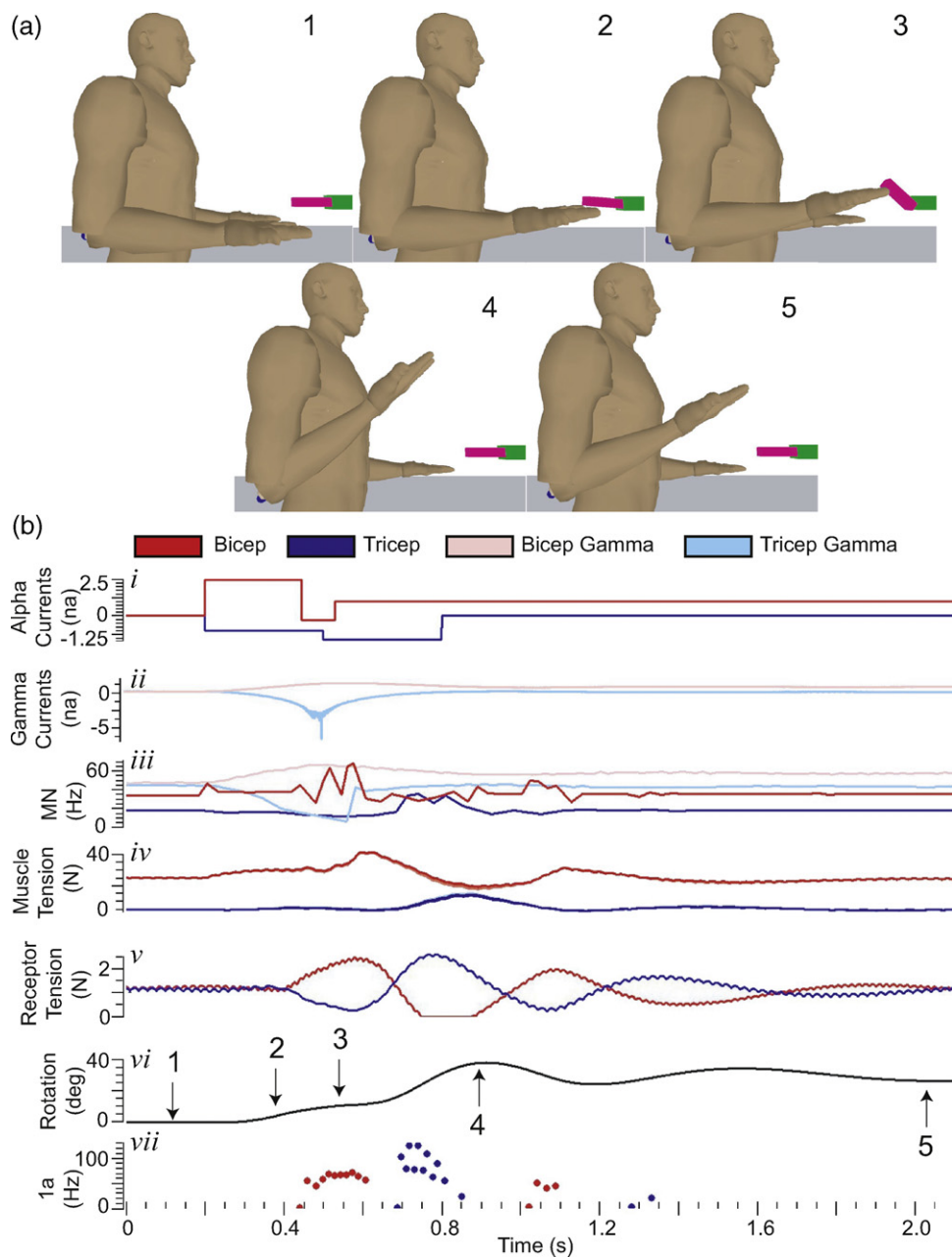


Fig. 5. Stretch reflex response to perturbation during arm flexion. (a) Simulator animation showing the arm flexing (1), then blocked (2, 3), overcoming the hinged block and overshooting its target position (4), and reaching its target position (5). (b) The italicized Roman numeral in each plot refers to the corresponding numbers in Fig. 3a. Stimulation of the MNs as in Fig. 4c caused the arm to flex quickly and the biceps spindle to shorten, thereby keeping its tension constant. When the hand struck the hinged block, the biceps spindle no longer shortened at the planned rate and the spindle tension increased, exciting the biceps 1a sensory neurons. The 1as excited the bicep alpha MN, and inhibited the triceps alpha MN, producing an increased flexion torque that enabled the arm to overcome the block. Overcompensation then dampened to leave the arm at the target location. The numbers in plot vi mark the times of the images in (a).

how the nervous system and body are ‘tuned’ to function synergistically within the animal’s habitat (Chiel et al., 2009). However, the modeling approaches are each tailored to a particular example of an animal’s behavior and the neural structures that mediate it. At present, no general simulator for neuromechanical mechanisms exists to provide the function that NEURON (Hines and Carnevale, 2001) and GENESIS (Bower and Beerman, 2007) provide as general neural simulators, or that OPENSIM (Delp et al., 2007) provides for biomechanical simulations. AnimatLab was created to provide such a general neuromechanical simulator.

AnimatLab provides a framework in which a wide variety of neuromechanical models can be assembled and analyzed in different specific contexts. The AnimatLab simulation presented here shows

how proprioceptive feedback can be kept sensitive to perturbation while being prevented from interfering with voluntary movements. Other AnimatLab simulations have shown how mechanical resonance in the cat leg enables a simple neural CPG to produce the rapid oscillatory movements of the paw shake (Klishko et al., 2008a,b), how the motor program that drives kicking in locust also can drive the jump (Cofer et al., in press), and how stable walking gaits in the crayfish result from mechanical coupling of the legs to each other through the body (Rinehart and Belanger, 2009).

In all of these simulations, the neuromechanical models allow questions to be addressed that cannot be approached directly by experiment. The major one of these questions is how an entire neural and mechanical system and its individual elements

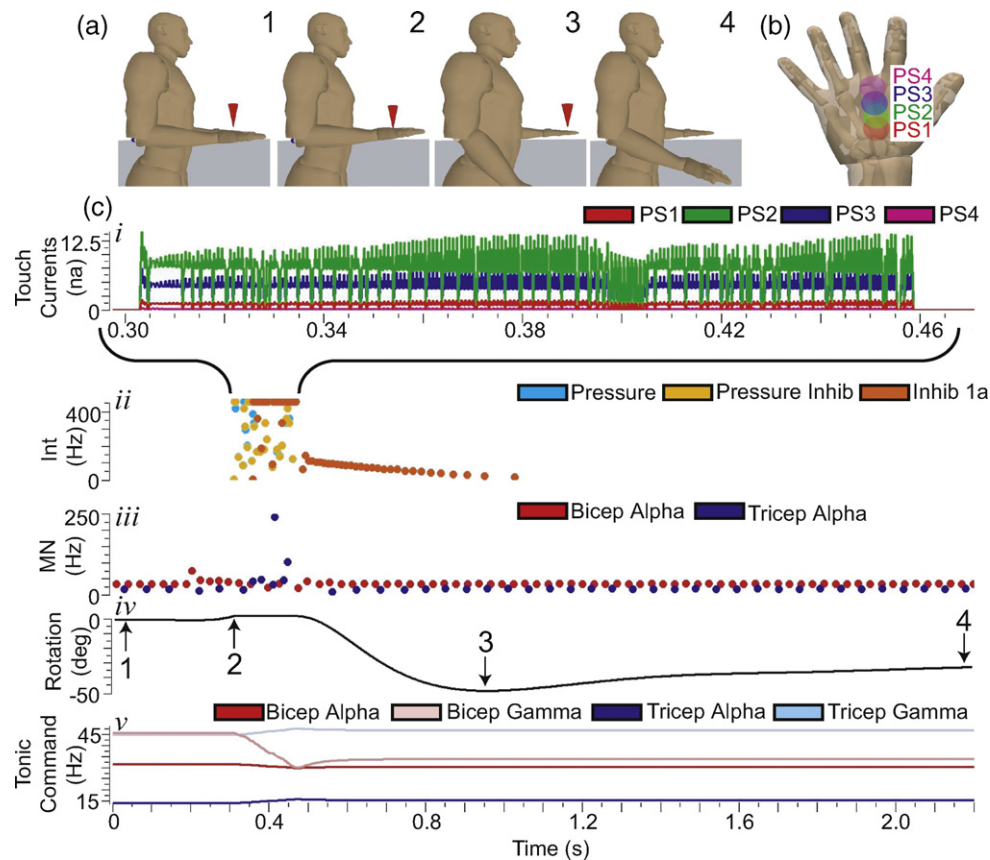


Fig. 6. Contact-withdrawal response to a sharp stimulus to the palm of the right hand. (a) Images from the animated model's response, showing the initial position of the right hand below the sharp red object (1), contact upon right arm flexion (2), extension of the right arm (3), and the new equilibrium position of the right arm (4). (b) The location of the four receptive fields PS1–4 on the hand. (c) Each graph axis corresponds to the italicized Roman numeral labels in Fig. 3b. Colors are coded to match the corresponding receptive field or neuron. (i) Generator currents in the sensory neurons during contact with the obstacle. (ii) Firing frequency response of the sensory interneurons. (iii) Alpha MN responses. (iv) Rotation of the arm; numbered arrows mark the times of occurrence of the frames in (a). (v) Reflex change in the tonic command to establish a new equilibrium arm position.

perform in a close-loop, dynamic, and unrestrained context. The individual neural circuit elements can usually only be monitored in anesthetized, restrained, and dissected preparations, where the conditions of the cells and circuits are quite different from those found in the intact, behaving animal. In simulation, the timing and amplitude relationships between motor neuron firing, muscle tension, limb movement, spindle tension, afferent responses, and motor neuron firing (again) can all become clear (Figs. 5 and 6).

AnimatLab models reflect specific hypotheses about how a system works and how it depends on the properties of individual elements. Simulations can test hypotheses by showing the circumstances under which the model behaves like the animal, identifying the model parameters that have more or less influence on that behavior, and revealing the effects of changes in parameter values that are not normally observable in the freely moving animal. Moreover, the sensitivities of the model to its organization and to its parameters provide another set of testable hypotheses about the corresponding elements of the real animal. This iterative dialog between experiment and simulation may allow the daunting complexity of dynamic neuromechanical systems to be analyzed and understood.

The AnimatLab models also provide a record of what is known or hypothesized about the individual elements, including the receptors, neurons, synapses, muscles, joints, and body segments, and about the organization of the entire neuromechanical system. Because information about the key elements is always incomplete, not all of the parameter values of simulated elements are deter-

mined by experiment; and values for those that are unknown must be estimated or guessed. Such provisional models are subject to frequent revision as they are tested and as parameter values are determined by new experiments. It is therefore important to keep a clear record of the source of each parameter value in the model so that what is known from measurement and what is estimated or guessed can be clearly distinguished. This is facilitated in AnimatLab by the “description” cell of each element’s “properties table”, where the sources of all the parameter values in the table can be identified either directly or through hyperlinks to other documents.

The arm flexion reflex simulations presented here demonstrate some of AnimatLab’s facilities for modeling and testing hypotheses of dynamic neuromechanical feedback and control. We demonstrated how appropriately scaled and timed tri-phasic alpha MN commands can produce rapid, smooth arm flexion movements (Fig. 4), and how gamma MN commands can provide a “plan” of the intended movement to enable myotactic reflexes to correct unexpected deviations from the plan produced by external perturbations (Fig. 5). In additional simulations not described here, we found that the command-evoked arm flexion and extension movements were sensitive to relatively modest changes in the amplitude and timing of the alpha and gamma motor commands. The final simulations demonstrated how spinal contact-avoidance reflexes have to be coupled with inhibition of myotactic reflexes and excitation of long-loop reflexes to allow the arm to move quickly to a new equilibrium point. A more formal model might be used to examine the sensitivity of the model’s behavior to different parameter values. These and other capabilities can be used to explore control

of multijoint or multi-limb movements in humans or any skeletal animal.

Beyond a single investigator's efforts to build and test models of a neuromechanical system is the need to share the results of those efforts with others. The lack of a common platform for building, testing and editing neuromechanical models has meant that they are not readily shared, and so has prevented them from being tested, modified or extended by others. By providing a common, flexible and broadly applicable modeling framework, AnimatLab enables investigators to share models, test them and modify them.

The next version of the software will include the ability to use deformable meshes so that soft-bodied animals and limbs (e.g., tongue and tentacle) can also be simulated. In addition, we plan to add new sensory systems, including vision and audition, and more sophisticated and realistic hydrodynamics to allow swimming behavior to be simulated. We also plan to enable users of the biomechanics simulator OPENSIM (Delp et al., 2007) and the neural simulator NEURON (Hines and Carnevale, 1997, 2001) to import models into AnimatLab to be extended for neuromechanical simulations. Finally, we plan to make AnimatLab a powerful tool for building and testing biomimetic robots by adding toolkits that interact with the Lego Mindstorms and Vex robotic kits. This will make it easy to use biologically realistic networks to control simple robots.

SIMM and OPENSIM are widely used general simulators for biomechanics, while NEURON and GENESIS play a similar role for neurophysiology. AnimatLab provides a general simulation tool for neuromechanics, at the intersection of biomechanics and neurophysiology, where no comparable general simulator exists. It is our hope that AnimatLab will prove as generally useful for this area as these other simulators have for theirs.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jneumeth.2010.01.005.

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